Uniform Douglas one-use, projected force, and low-completion boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

R-143 put the complete fixed-chart action into the one-use feature form

$$\mathcal{Y}_\pi(h) = \Phi_\pi(h) \oplus \sqrt{\frac{9}{10}}\, h \oplus \sqrt{\frac{3}{10}}\, |Z_{K,h}|^2 Z_{K,h}, \qquad 2\mathcal{A}_\pi(h) = \|\mathcal{Y}_\pi(h)\|^2 - \|\mathcal{U}_\pi(h)\|^2. \tag{1.1}$$

The missing production contraction was previously phrased as an exact map from $\mathcal{Y}$ to $\mathcal{U}$. The first new result of R-144 is that exactness is not necessary. A strict common-output contraction plus a uniformly bounded returned-low residual already gives both strict one-use coefficients, with a sharp residual constant. It is therefore one sufficient route to T-050 and does not require second differentiation of every production feature.

The second result separates that route from the projected-force Hessian alternative. A positive reduced/source-low effective Hessian gap improves the 9/20 source coefficient, while the retained sextic loss 3/20 is already strictly below the canonical T-050 threshold $\gamma/6 = 27/100$. Thus the reduced effective gap, origin-force bound, and absolute anchor form a second sufficient route. This comparison was corrected during hostile pre-registration review: the first draft incorrectly compared 3/20 with itself rather than with $\gamma/6$.

The third result makes the remaining matrix request type-correct and falsifiable. The Hessian domain is the declared predictable source chart plus genuine low auxiliaries. The q, fibre, reveal, and visit labels enter that domain only when the chart declares independent tangent directions; otherwise they label maps into a common physical feature space. Exact phase-cycle, returned-low, and equal-first-jet fixtures prove that the current local data do not determine the production sign.

All conclusions are at fixed finite cutoff and positive floor, except for explicitly uniform hypotheses. No production chart, common-output contraction, bounded residual, absolute anchor, positive Hessian gap, origin-force bound, T-050 closure, A13 gate, Nelson estimate, removal, interacting measure, Sector-A closure, or tier promotion is asserted.

2. Sharp affine-residual complete-feature theorem

Let the owner-complete action, including its absolute normalization, be

$$\mathcal{A}_{J,\pi}(h) = a_{J,\pi} + \frac{1}{2}\{\|\mathcal{Y}_{J,\pi}(h)\|^2 - \|\mathcal{U}_{J,\pi}(h)\|^2\}. \tag{2.1}$$

The feature $\mathcal{Y}$ is exactly (1.1): its source and sextic coordinates are already present and may not be paid again. Assume that, on the complete production feature span, there is one linear common-output map $C_{J,\pi}$ and residual $E_{J,\pi}(h)$ such that

$$\mathcal{U}_{J,\pi}(h) = C_{J,\pi}\mathcal{Y}_{J,\pi}(h) + E_{J,\pi}(h). \tag{2.2}$$

Theorem 2.1 (sharp affine-residual Douglas bound). Suppose there are constants

$$0 \le \rho < \sigma < 1, \qquad M < \infty, \qquad C_0 < \infty \tag{2.3}$$

uniform in cutoff, chart, refinement, and admissible h, with

$$\|C_{J,\pi}\| \le \rho, \qquad \|E_{J,\pi}(h)\| \le M, \qquad a_{J,\pi} \ge -C_0. \tag{2.4}$$

Then

$$\boxed{\mathcal{A}_{J,\pi}(h) \ge -C_0 - \frac{\sigma^2 M^2}{2(\sigma^2-\rho^2)} + \frac{1-\sigma^2}{2}\|\mathcal{Y}_{J,\pi}(h)\|^2.} \tag{2.5}$$

The residual coefficient in (2.5) is sharp under norm data alone.

Proof. Put $y = \|\mathcal{Y}\|$ and $m = \|E\|$. The triangle inequality and $\|C\| \le \rho$ give $\|\mathcal{U}\| \le \rho y + m$. For $\delta = \sigma^2 - \rho^2 > 0$,

$$\begin{aligned} &\sigma^2 y^2 + \frac{\sigma^2}{\delta} m^2 - (\rho y + m)^2 \\ &\qquad = \frac{1}{\delta}(\delta y - \rho m)^2 \ge 0. \end{aligned} \tag{2.6}$$

Substitution in (2.1) proves (2.5). Equality in (2.6) occurs at $y = \rho m/\delta$, proving sharpness. ☐

Write

$$X(h) = \mathbb{E}\sum_k \|h_k\|_2^2, \qquad Y_6(h) = \mathbb{E}\|Z_{K,h}\|_6^6. \tag{2.7}$$

Since

$$\|\mathcal{Y}(h)\|^2 = \|\Phi(h)\|^2 + \frac{9}{10}X(h) + \frac{3}{10}Y_6(h), \tag{2.8}$$

subtracting the stabilizers from (2.5) gives the exact consequence

$$\boxed{\begin{aligned} \mathbb{E}V_J^{\mathrm{ren}}(Z_h) \ge\ & -\frac{9}{20}\sigma^2 X(h) - \frac{3}{20}\sigma^2 Y_6(h) \\ & + \frac{1-\sigma^2}{2}\|\Phi(h)\|^2 - C_0 - \frac{\sigma^2 M^2}{2(\sigma^2-\rho^2)}. \end{aligned}} \tag{2.9}$$

Choose the comparison exponent $p = 11/10$ and retain the Nelson exponent $q = 10/9$. Then $q - p = 1/90$ and

$$\epsilon_v = \frac{9}{20}\sigma^2 < \frac{9}{20} = \frac{1}{2q} < \frac{5}{11} = \frac{1}{2p}, \qquad \frac{5}{11} - \frac{9}{20} = \frac{1}{220}, \qquad \epsilon_6 = \frac{3}{20}\sigma^2 < \frac{3}{20} < \frac{\gamma}{6}. \tag{2.10}$$

where the A1 value is $\gamma = 81/50$ and $\gamma/6 = 27/100$.

Corollary 2.2. The uniform hypotheses (2.1)–(2.4), on the full R-093 directed chart union, are sufficient for the controlled-shell one-use inequality T-050. They have not been proved for the production features. If $E = 0$, one may let $\sigma \downarrow \rho$ and recover the exact R-143 contraction coefficients.

The bounded residual is useful for the actual research interface: a returned-low column need not vanish or be absorbed into a second source payment. It may remain as E, provided its complete Hilbert norm has a cutoff/chart-uniform bound. A residual which grows with h or the cutoff does not meet (2.4).

3. Global feature route versus Hessian fallback

The exact projected production force from R-141 is

$$\left(g_\pi^{\mathrm{prod}}(h)\right)_b = \frac{9}{10}h_b + \mathbb{E}\left[S_b^*\left\{\frac{9}{10}|Z_h|^4 Z_h - \frac{1}{2}\sum_\alpha G_{\mathcal{T},\alpha}(Z_h)\right\}\middle|\mathcal{F}_{t_{b-1}}\right]. \tag{3.1}$$

The covariance-normal future-variance force cancels when the actual action owner is restored and is not appended again.

If i, j are declared source/low chart directions, the action-signed Hessian is

$$\begin{aligned} H_{ij} = {} & \frac{9}{10} G^{\rm src}_{ij} + \Re\langle \Phi_i, \Phi_j \rangle + \Re\langle \Phi, \Phi_{ij} \rangle \\ & - \Re\langle U_i, U_j \rangle - \Re\langle U, U_{ij} \rangle + S^{(6)}_{ij}. \end{aligned} \tag{3.2}$$

There is no extra factor two in (3.2). If a computation starts from the raw signature $K = \|U\|^2 - \|\Phi\|^2$, it must first use

$$D^2 \mathcal{P}_{\rm comp} = -\frac{1}{2} D^2 K. \tag{3.3}$$

Theorem 3.1 (source-gap sufficient condition). Let $\mathcal{A}^{\rm red}_{J,\pi}(h)$ be a C^2 complete action on a source domain star-shaped about zero. It must equal the action pulled back to the exact admissible low graph, including every graph connection term. If the low is an independent variable eliminated stationarily, it is instead the attained C^2 reduced action after the null-range and regularity hypotheses that make the stationary reduction attained and C^2, not the bare source–source principal block. Write $g^{\rm red} = D\mathcal{A}^{\rm red}$. Equip every declared source chart with $\|h\|^2_{\rm src} = X(h)$ and use its realified six-real Hilbert pairing. Assume that the same constants $\mu > 0$, $B_g < \infty$, and $C_0 < \infty$ work uniformly in cutoff, chart, representation-preserving refinement, and admissible control. For every declared source direction v and every $t \in [0, 1]$, require

$$D^2 \mathcal{A}^{\rm red}(th)[v, v] \geq \mu \|v\|^2_{\rm src}, \tag{3.4}$$

and

$$\|g^{\rm red}(0)\|^2_{{\rm src},*} \leq B_g, \qquad \mathcal{A}^{\rm red}(0) \geq -C_0. \tag{3.5}$$

Then for every $0 < \theta < 1$,

$$\boxed{\mathbb{E} V^{\rm ren}_J(Z_h) \geq -\left\{ \frac{9}{20} - \frac{(1-\theta)\mu}{2} \right\} X(h) - \frac{3}{20} Y_6(h) - C_0 - \frac{B_g}{2\theta\mu}.} \tag{3.6}$$

All quantities in (3.6), including Z_h and $Y_6(h)$, are evaluated on the same reduced low graph used to define $\mathcal{A}^{\rm red}$.

Proof. Strong convexity gives $\mathcal{A}^{\rm red}(h) \geq \mathcal{A}^{\rm red}(0) + \Re\langle g^{\rm red}(0), h \rangle + \mu X(h)/2$. The dual Young bound

$$\Re\langle g^{\rm red}(0), h \rangle \geq -\frac{\theta\mu}{2} X(h) - \frac{B_g}{2\theta\mu} \tag{3.7}$$

and subtraction of the two stabilizers prove (3.6). □

Equation (3.6) gives a strict source coefficient and the sextic coefficient 3/20. For $p = 11/10$, the source comparison is

$$\max\left\{0, \frac{9}{20} - \frac{(1-\theta)\mu}{2}\right\} < \frac{9}{20} < \frac{5}{11} = \frac{1}{2p}, \qquad \frac{5}{11} - \frac{9}{20} = \frac{1}{220}. \tag{3.8}$$

The canonical sextic comparison is

$$\frac{\gamma}{6} = \frac{27}{100}, \qquad \frac{3}{20} < \frac{27}{100}, \qquad \frac{27}{100} - \frac{3}{20} = \frac{3}{25}. \tag{3.9}$$

Consequently (3.4)–(3.5), uniformly on the full directed chart union, are already sufficient for T-050. No additional sextic reserve is required. If the displayed source coefficient is negative, discard the favorable positive multiple of X and take

$$\epsilon_v = \max\left\{0, \frac{9}{20} - \frac{(1-\theta)\mu}{2}\right\} < \frac{9}{20} < \frac{5}{11} = \frac{1}{2p}, \qquad \epsilon_6 = \frac{3}{20} < \frac{27}{100}. \tag{3.10}$$

The unproved production inputs are the positive reduced/source-low Feshbach gap, the origin-force bound, and the absolute anchor.

This comparison leaves two efficient sufficient routes. The global theorem in Section 2 creates both margins and makes second jets optional. The local route (3.2) is equally valid if the common-output contraction fails or cannot be proved, provided its reduced source gap, origin force, and anchor are uniform.

4. Type-correct source, packet, and low domains

For a fixed temporally faithful chart,

$$\mathcal{S}_\pi^{\mathrm{pred}} = \bigoplus_b L^2(\Omega, \mathcal{F}_{t_{b-1}}; \mathcal{S}_b), \qquad L_\pi h = \sum_b S_b h_b. \tag{4.1}$$

Let ℓ be a genuinely independent returned-low coordinate and write the physical positive feature as

$$P(x,\ell) = \Phi(x,\ell) \oplus \sqrt{\frac{3}{10}}|Z(x,\ell)|^2 Z(x,\ell). \tag{4.2}$$

With

$$E(x,\ell) = \frac{1}{2}\{\|P(x,\ell)\|^2 - \|U(x,\ell)\|^2\}, \qquad T = \mathrm{diag}(L_\pi, \mathrm{I}_\ell), \tag{4.3}$$

the full action is

$$\mathcal{A}(h,\ell) = \frac{9}{20}\|h\|^2 + E(T(h,\ell)), \tag{4.4}$$

and hence

$$\boxed{D^2\mathcal{A} = \begin{pmatrix} \frac{9}{10}\mathrm{I} & 0 \\ 0 & 0 \end{pmatrix} + T^*(D^2E)T.} \tag{4.5}$$

In physical/low blocks $Q = D^2E$, this means

$$\begin{gathered} H_{hh} = \frac{9}{10}\mathrm{I} + L_\pi^* Q_{xx} L_\pi, H_{h\ell} = L_\pi^* Q_{x\ell}, \\ H_{\ell h} = Q_{\ell x} L_\pi, H_{\ell\ell} = Q_{\ell\ell}. \end{gathered} \tag{4.6}$$

For $n \in \mathrm{Ker}\, L_\pi$,

$$D^2\mathcal{A}[(n,0),(m,k)] = \frac{9}{10}\Re\langle n, m\rangle. \tag{4.7}$$

Thus $\mathrm{Ker}\, L_\pi$ carries exactly $9\mathrm{I}/10$ and decouples after the orthogonal split

$$\mathcal{S}_\pi = \mathrm{Ker}\, L_\pi \oplus (\mathrm{Ker}\, L_\pi)^\perp. \tag{4.8}$$

It is a subspace, not an extra coordinate. If instead the low is graph-tied, $\ell = \Lambda h$, the synthesis is (L_π, Λ); (4.7) requires $\Lambda(\mathrm{Ker}\, L_\pi) = 0$. A nonlinear Λ also has connection terms.

For the R-143 $q = (5,6,7)$ family, three gaps times four active fibre modes give a 12-coordinate local stencil only after a chart declares those tangent directions. Adding the two phase directions gives 18 fibre coordinates per declared temporal coordinate. The R-142 two-root/two-visit scalar chart has three source coordinates, so a formal tensor chart would have 54 packet coordinates, 36 active and 18 phase. This is not a production dimension: the tensor independence and cross blocks have not been registered.

No canonical production temporal index set is registered here. The future chart must define every root, visit, reveal time, source atom, and filtration constraint explicitly, after same-root visits have been telescoped before owner formation. Its q/channel/reveal pieces may be maps from that declared source domain into one common output. Replacing the common output by an artificial direct sum deletes their cross Gram and changes the form. This is a specification requirement, not an asserted production coordinate formula.

5. Endpoint secants and refinement

For two complete endpoints $q = (h, \ell)$ and $q_0 = (h_0, \ell_0)$, exact polarization gives

$$\begin{aligned} 2\{\mathcal{A}(q) - \mathcal{A}(q_0)\} = {} & \frac{9}{10}(\|h\|^2 - \|h_0\|^2) \\ & + \Re\langle P(q) + P(q_0), P(q) - P(q_0)\rangle \\ & - \Re\langle U(q) + U(q_0), U(q) - U(q_0)\rangle. \end{aligned} \tag{5.1}$$

This is an exact derivative-free endpoint route and retains baseline/low crosses. It can replace the jet route when a uniform global contraction such as Theorem 2.1 is proved.

It must not be confused with a finite-step local Hessian. Define the symmetric difference by $\Delta_{\varepsilon v}E(x) = E(x + \varepsilon v) - E(x - \varepsilon v)$. For a smooth functional E,

$$\frac{\Delta_{\varepsilon v}\Delta_{\eta w}E(x)}{4\varepsilon\eta} = \frac{1}{4\varepsilon\eta}\int_{-\varepsilon}^{\varepsilon}\int_{-\eta}^{\eta} D^2E(x + sv + tw)[v, w]\,dt\,ds. \tag{5.2}$$

Only a limit or a certified Hessian-variation remainder turns (5.2) into the local value at x.

If a representation-preserving refinement has a linear isometric source embedding ι and $\mathcal{A}_\pi = \mathcal{A}_{\pi'} \circ I$, then

$$D^2\mathcal{A}_\pi = I^* D^2\mathcal{A}_{\pi'} I. \tag{5.3}$$

Without isometry the source-cost defect is

$$\frac{9}{10}(\mathrm{I} - \iota^*\iota). \tag{5.4}$$

A nonlinear refinement has the additional chain-rule connection $D\mathcal{A}_{\pi'}\, D^2 I$. Congruence, not equality of matrix spectra, is the invariant statement.

6. Exact q567 phase-cycle non-identifiability

Let B be the exact positive rank-four R-142 active fibre Gram at its registered audit point. The executable reconstruction uses the R-142 Schur coefficient

$$a = \frac{c_{JJ}\alpha_X^2}{P}, \quad b = \frac{c_{JK}\alpha_X\beta_X}{P}, \quad c = \frac{c_{KK}\beta_X^2}{P}, \quad c_0 = a - \frac{b^2}{c} = \frac{3}{250P}, \tag{6.0}$$

not the uncompleted diagonal coefficient a. A hostile pre-registration audit caught and corrected that omitted Schur term before release. Consider

$$L_+ = \begin{pmatrix} 1 & 3/4 & 3/4 \\ 3/4 & 1 & 3/4 \\ 3/4 & 3/4 & 1 \end{pmatrix}, \qquad L_- = \begin{pmatrix} 1 & 3/4 & 3/4 \\ 3/4 & 1 & -3/4 \\ 3/4 & -3/4 & 1 \end{pmatrix}. \tag{6.1}$$

They have the same diagonal and the same magnitude on every edge, but

$$\operatorname{spec} L_+ = \{5/2, 1/4, 1/4\}, \qquad \operatorname{spec} L_- = \{7/4, 7/4, -1/2\}. \tag{6.2}$$

Their determinants are $5/32$ and $-49/32$. The products of the three edge signs are $27/64$ and $-27/64$. Under layer sign gauges $s_{ij} \mapsto d_i d_j s_{ij}$ this cycle product is invariant, so the two completions are not gauge-equivalent. Consequently

$$\operatorname{inertia}(L_+ \otimes B) = (12, 0, 0), \qquad \operatorname{inertia}(L_- \otimes B) = (8, 4, 0). \tag{6.3}$$

There is an even smaller mixed-only obstruction. With

$$G_Y = \mathrm{I}_3, \qquad G_U = \frac{3}{8}\mathbf{1}\mathbf{1}^T, \tag{6.4}$$

every diagonal margin is $5/8 > 0$, but

$$\operatorname{spec}(G_Y - G_U) = \{1, 1, -1/8\}, \qquad \rho_*^2 = \frac{9}{8}. \tag{6.5}$$

Tensoring with B yields four adverse active-fibre directions. These are information-theoretic completions, not production counterdirections. They prove that diagonal fibre data and edge magnitudes cannot decide the signed common-output block.

7. Returned low and legal reverse

Even a fully known positive 12-coordinate core does not decide the augmented low sign. Embed

$$A_{12} = \begin{pmatrix} 1 & 29/32 \\ 29/32 & 1 \end{pmatrix} \oplus \mathrm{I}_{10}, \qquad \det A_{12}^{(2)} = \frac{183}{1024} > 0. \tag{7.1}$$

Couple one returned-low coordinate by

$$v_+ = (1/4, 1/4, 0, \dots, 0), \qquad v_- = (1/4, -1/4, 0, \dots, 0), \tag{7.2}$$

and give it unit diagonal. The two 13-coordinate completions have identical core, diagonals, and entrywise magnitudes, while

$$\det H_+ = \frac{171}{1024} > 0, \qquad \det H_- = -\frac{61}{1024} < 0. \tag{7.3}$$

This is the exact R-141 signed-low fixture embedded in the proposed core.

Forward and legal reverse are one self-adjoint off-diagonal owner. The entry 29/32 in (7.1) already represents their symmetric block. Charging it a second time replaces it by 29/16 and mutates the determinant to

$$1 - (29/16)^2 = -\frac{585}{256}. \tag{7.4}$$

Thus the unified matrix must contain the adjoint once, not assign a second budget to "reverse." Balanced and low are other blocks of the same form, not independent opportunities to reuse source or sextic reserves.

8. Base and first jets do not determine the Hessian

Let

$$U(t) = t, \qquad \Phi_\pm(t) = 1 + t \pm t^2, \qquad K_\pm(t) = U(t)^2 - \Phi_\pm(t)^2. \tag{8.1}$$

The two features have the same base and first jet. Nevertheless

$$K_+''(0) = -4, \qquad K_-''(0) = 4, \tag{8.2}$$

and the action owner $\mathcal{P}_{\mathrm{comp}} = -K/2$ has

$$\mathcal{P}_+''(0) = 2, \qquad \mathcal{P}_-''(0) = -2. \tag{8.3}$$

After the source-stabilizer Hessian 9/10 is added, the signs remain opposite:

$$H_{\mathcal{A},+} = \frac{29}{10}, \qquad H_{\mathcal{A},-} = -\frac{11}{10}. \tag{8.4}$$

The unit signed feature-chord quadratic values are 3 and -1, not the local Hessians 2 and -2. Hence a finite endpoint chord is not a substitute for the connection terms $\langle \Phi, \Phi_{ij}\rangle - \langle U, U_{ij}\rangle$. The alternatives are precise: prove the global feature theorem of Section 2, or supply the full cross-second jets and use (3.2).

Sections 6–8 are registered together as `NG-2026-08-02-A13-LOCAL-STENCIL-PRODUCTION-SIGN-NONIDENTIFIABILITY`. The no-go rejects an inference from incomplete data; it does not reject the production action or T-050.

9. Unified production decision theorem

The current proof tree now has two nonduplicative branches.

Branch A: global complete-feature route. Register one canonical temporal chart and common-output owner map, including the conditional/complete low distinction and the shared covariance probe. Prove (2.2)–(2.4) uniformly. This directly gives (2.9), both strict margins, and T-050. No U/Φ second jets are needed on this branch.

Branch B: local action route. If Branch A fails or is unavailable, register

$$Y_0, U_0, \qquad Y_i, U_i, \qquad Y_{ij}, U_{ij} \tag{9.1}$$

with the same future projection, output cluster, owner partition, and shared probe. Assemble (3.2), with the exact anisotropic finite-low and returned-low blocks. For stationary elimination, type H_{00} as the eliminated low/danger block and H_{11} as the source block. Uniformly at every point of the reduced graph and for every cutoff, cylindrical chart, and representation-preserving refinement, check

$$H_{00} \succeq 0, \qquad \operatorname{Ran} H_{01} \subseteq \operatorname{Ran} H_{00}^{1/2}, \tag{9.2}$$

and let W be the Douglas reduced solution

$$H_{00}^{1/2} W = H_{01}, \qquad H_{\text{eff}} = H_{11} - W^* W. \tag{9.3}$$

This is the bounded-Hilbert-space reduced-solution form of the exact R-140 finite-dimensional graph–Feshbach algebra. On a finite cylindrical chart it agrees with $H_{11} - H_{10} H_{00}^{\dagger} H_{01}$; no unbounded pseudoinverse is used on the full predictable L^2 source space. Require uniformly $H_{\text{eff}} \succeq \mu_{\text{finite}} G_{\text{src}}$. To feed Theorem 3.1, the stationary infimum must define the same C^2 reduced action (for example through an attained minimizer with a regular stationary graph), and the uniform cylindrical estimates must extend to the declared source domain. Add the transformed analytic tail only after the finite block. Here H_{tail} must already be the reduced/pulled-back source tail, rather than an unevaluated ambient mixed block. Prove one uniform $\mu = \mu_{\text{finite}} - \|H_{\text{tail}}\| > 0$, so that, relative to the source metric G_{src} defining $X(h)$,

$$H_{\text{eff}} + H_{\text{tail}} \succeq \mu G_{\text{src}}. \tag{9.4}$$

on the entire reduced graph. If low is instead a prescribed nonlinear graph, replace (9.3) by the full pullback Hessian, including the $D\mathcal{A}\, D^2\Lambda$ connection term. In either case the force bound and anchor in (3.5) must be those of the same reduced action. These hypotheses give (3.4), hence T-050 because both $9/20 < 5/11$ and $3/20 < \gamma/6$. A nonpositive edge must be lifted through (9.3) and checked as a legal action direction; it is not enough that a Douglas feature test failed.

The minimal machine-readable production object is therefore:

1. the reveal order, visits, source blocks, bases, and synthesis L_π;
2. full q567 Fourier vectors, real-even normalization, phases, and common output-cluster assignments;
3. one owner roster for covariance root, future projection, heat coordinate, output, channel, and shared trace probe;

4. either the complete endpoint maps $(\mathcal{Y}, \mathcal{U})$ on the admissible graph, or every base, first, and cross-second jet;
5. independent conditional-low and complete returned-low maps, or the exact graph law tying them to h;
6. the anisotropic finite-low block, complement gap, transformed tail, and representation-preserving refinement embeddings;
7. a uniform contraction/residual/anchor certificate for Branch A, or a Hessian/Feshbach/origin-force/anchor certificate for Branch B.

This unifies forward, legal reverse, balanced, and low. They are block labels inside one self-adjoint complete-owner object. Primitive trace, future variance, R-063 forest, predictable mean, and returned low remain alternate or signed coordinates with their registered multiplicities; none becomes a new positive reserve merely by appearing in the matrix.

10. Route ledger and consequences

Route	Verdict	Consequence
Exact global contraction $\mathcal{U} = C\mathcal{Y}$	conditional advance	R-143 criterion; R-144 translates it to both strict T-050 coefficients.
Strict contraction plus bounded low residual	new conditional advance	Theorem 2.1; exact low cancellation is unnecessary, and the residual cost is sharp.
Positive reduced/source-low effective Hessian gap with force and anchor	conditional advance	Improves ϵ_v and retains $\epsilon_6 = 3/20 < 27/100$ with margin $3/25$.
Draft sextic/source threshold comparison	corrected audit	T-050 uses $\epsilon_6 < \gamma/6$ and $\epsilon_v < 1/(2p)$, not an extra $3/20$ sextic reserve.
Uncompleted fibre coefficient a	corrected audit	The registered R-142 fibre uses $c_0 = a - b^2/c = 3/(250P)$.
Finite endpoint secant equals local Hessian	rejected	It is an averaged Hessian; use a limit/remainder or the global feature route.
Treat 12 q567 active entries as the full production matrix	rejected	Temporal, phase, signed common-output, anisotropic-low, and returned-low data remain undeclared.
Infer sign from diagonal/magnitude data	rejected	Opposite phase-cycle and returned-low completions preserve those data.
Count legal reverse separately	rejected	It doubles one self-adjoint owner and can flip the verdict spuriously.
Use base and first jets only	rejected	The exact $\Phi_{\pm}$ fixture has opposite true Hessians.

The two hostile corrections are registered as `AUDIT-2026-08-02-A13-R144-SEXTIC-THRESHOLD-CORRECTION` and `AUDIT-2026-08-02-A13-R144-FIBRE-SCHUR-COEFFICIENT-CORRECTION`; their append-only route corrections are `EXP-000601` and `EXP-000602`.

The result is genuine forward progress but not a numerical production verdict. It removes two unnecessary burdens from the main route: exact low residual vanishing and mandatory second jets. It also exposes the Hessian route as a second sufficient condition after correcting the sextic-threshold comparison. What remains is not an unspecified larger matrix; it is the declared common-output/low object in Section 9.

11. Devil's-advocate review

1. **Objection: Theorem 2.1 proves the production contraction. UPHELD against that claim.** It is a sharp implication from uniform production hypotheses. No such C, M, or C_0 is registered.

2. **Objection: a bounded residual can be any omitted owner. UPHELD.** The residual must be the exact difference $\mathcal{U} - C\mathcal{Y}$ after complete owner recombination, and its Hilbert norm must be uniform. A separately estimated duplicate is illegal.

3. **Objection: reduced source convexity closes T-050. VALID WITH MITIGATION.** Uniform positive reduced source convexity closes the source coefficient, and $3/20 < \gamma/6$ already closes the sextic coefficient. The production gap, origin-force bound, and absolute anchor are still absent.

4. **Objection: the source-null direction always decouples from low. VALID WITH MITIGATION.** It decouples for an independent low after the orthogonal kernel split. A graph-tied low requires $\Lambda(\operatorname{Ker} L_\pi) = 0$, and nonlinear graphs add connection terms.

5. **Objection: endpoint polarization eliminates second jets in all local calculations. UPHELD.** It eliminates them for a global feature inequality or exact endpoint comparison, not for a finite-step local Hessian without a remainder estimate.

6. **Objection: the adverse phase-cycle and low fixtures refute production positivity. UPHELD.** They prove non-identifiability from the currently registered partial data only. They are not legal production directions.

7. **Objection: 12 active q567 channels determine the source dimension. UPHELD.** They determine a local stencil only after a chart declares independent source tangents; common-output feature labels can share one smaller domain.

8. **Objection: R-144 closes A13 or Sector A. DISMISSED.** Both A13 gates, T-050, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removals, the interacting measure, and Sector A remain open at T4.

12. Result footer

The primary certificate derives the A1 fibre coefficients from the production manifest, proves the sharp residual square symbolically, checks both one-use coefficients, the Hessian T-050 margin, source/low pullback, endpoint polarization, central-secant limit, refinement congruence, actual rank-four fibre phase-cycle inertia, returned-low completions, legal-reverse factor, and second-jet fixture. Run

```
$script = "codes/foundations/" +
 "a13_classii_uniform_douglas_one_use_" +
 "projected_force_low_completion_boundary.py"
E:\Dev\TECT.venv\Scripts\python.exe $script
```

The independent certificate uses only the Python standard library and exact `Fraction` arithmetic. Run

```
$script = "codes/foundations/" +
 "a13_classii_uniform_douglas_one_use_" +
 "projected_force_low_completion_boundary_independent.py"
E:\Dev\TECT.venv\Scripts\python.exe $script
```

The expected outputs are respectively `R-144 primary: 61/61 PASS` and `R-144 independent: 56/56 PASS`. The integrated verifier reruns both certificates, pins every authority including the canonical T-050 gate, checks the negative and exploration ledgers and public surfaces, rebuilds and renders this PDF deterministically, and enforces the no-overclaim flags.

Result ID: A13-CLASSII-UNIFORM-DOUGLAS-ONE-USE-PROJECTED-FORCE-LOW-COMPLETION-BOUNDARY.
Ledger ID: R-144.
Precise statement: sharp affine-residual complete-feature implication; alternative C2 reduced-action Hessian implication; exact phase-cycle, returned-low, legal-reverse, and equal-first-jet information boundaries.
Scope: fixed finite cutoff and positive floor for exact chart identities; uniform only under the explicit hypotheses of Theorem 2.1 or Theorem 3.1.
Dependencies: A1; R-093, R-104, R-130, R-140--R-143; claims/GATES.md.
Evidence grade: ANALYTIC, EXACT, CONDITIONAL, EXECUTED, AUDITED.
Audit status: two hostile corrections registered with EXP-000593--EXP-000602.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_uniform_douglas_one_use_projected_force_low_completion_boundary_verify.py
Expected output: integrated 313/313 PASS after primary 61/61 and non-importing independent 56/56, exact cross-checks, deterministic PDF rebuild, exploration/negative/authority/manifest, all-page render, and generated public-surface checks pass.
Falsification gate: any failed sharp-square identity, T-050 threshold bridge, reduced-action hypothesis, exact fixture, authority/hash, deterministic PDF, public-surface, or no-overclaim check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no canonical production chart or uniform Branch-A or Branch-B hypotheses, legal adverse production direction, T-050 closure, A13 closure, Nelson theorem, interacting measure, or Sector-A closure follows.
Next required action: register the canonical temporal/common-output/conditional-low/returned-low chart; test Branch A first and Branch B on the same reduced action if Branch A is unavailable.